

Question	Answer	Marks
3(a)(i)	N : number of molecules (of the gas)	B1
	m : mass of one molecule (of the gas)	B1
	$\langle c^2 \rangle$: mean square speed (of molecules)	B1
3(a)(ii)	$pV = NkT$	M1
	$NkT = \frac{1}{3}Nm\langle c^2 \rangle$ and $E_K = \frac{1}{2}m\langle c^2 \rangle$ leading to $E_K = (3/2) kT$	A1
3(b)	$\frac{1}{2} \times 3.34 \times 10^{-27} \times 9300^2 = (3/2) \times 1.38 \times 10^{-23} \times T$	C1
	$T = 6980 \text{ K}$	A1
3(c)(i)	$L = F \times 4\pi d^2$	C1
	$L = 2.52 \times 10^{-8} \times 4\pi \times (4.16 \times 10^{16})^2$ $= 5.48 \times 10^{26} \text{ W}$	A1
3(c)(ii)	$L = 4\pi\sigma r^2 T^4$ $5.48 \times 10^{26} = 4\pi \times 5.67 \times 10^{-8} \times r^2 \times 6980^4$	C1
	$r = 5.69 \times 10^8 \text{ m}$	A1
3(d)	(very high pressure so) molecules are (very) close together (<i>not just 'nearer'</i>)	B1
	forces between molecules are not negligible or volume of molecules not negligible compared with gas volume	B1

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Question	Answer	Marks
1(a)	(motion in which) acceleration is (directly) proportional to displacement	D1